THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An electric kettle and an electric bell are examples of which two effects respectively? 1

(a) chemical and heating **(b)** magnetic and heating
(c) heating and magnetic **(d)** magnetic and chemical

2. Assertion (A): Wires carrying heavy loads are made thick. Reason (R): Thick wires have lower resistance and heat less. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. An immersion rod and a table lamp are connected in parallel. If the lamp's filament breaks, the rod: 1

(a) stops **(b)** shocks
(c) continues to heat **(d)** becomes a magnet

4. Why is the earth wire connected to the metal body of an appliance? 1

(a) to make it a magnet
(b) to increase current
(c) so leakage current flows to the ground and not through the user
(d) to heat it

5. Assertion (A): A short circuit is dangerous. Reason (R): It causes a very large current to flow. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A circuit carries a normal load of 12 A. The most suitable fuse rating is: 1

(a) 10 A **(b)** 5 A **(c)** 100 A **(d)** 15 A

7. The magnetic field around a straight current-carrying wire is: 1

(a) in circles around the wire **(b)** zero
(c) along the wire **(d)** pointing outwards

8. Assertion (A): Wet hands should not touch electrical switches. Reason (R): Water, with dissolved salts, is a good conductor. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Increasing the number of turns of a coil (with the same current) makes an electromagnet stronger. 1

Reason (R): A coil of insulated copper wire is used to make an electromagnet.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** An electric heater coil is made of nichrome. 1

Reason (R): Nichrome is an alloy with high resistance and a high melting point, so it can get red hot without melting.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is it unsafe to use a coin in place of a fuse? 2

12. Why is an electromagnet with soft iron better than one with steel for an electric bell? 2

13. State why metal body appliances use a three-pin plug. 2

14. Identify the effect used: 2

(a) toaster

(b) lifting of iron by a crane

(c) electric bell.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A 3-pin plug has a long earth pin. Explain why it is longer than the others. 3

16. A factory electromagnet must be released quickly. Why is soft iron used and not steel? 3

17. Justify: 'An electromagnet is a temporary magnet whose strength can be controlled.' 3

OR

Why should a fuse be placed on the live wire only and not on the neutral? 3

18. Ravi says 'electricity in a cell is the same as the electricity at home'. Compare the two. 3

OR

Describe three precautions when using an MCB or fuse in the house. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sana places a compass needle below a straight wire running north to south, with the needle parallel to the wire. When she connects the wire to a battery through a switch, the needle swings. She then reverses the battery connections and the needle swings the other way. 5

(i) Who first observed this effect?

(ii) Name the effect of current shown.

(iii) What happens when the switch is kept open?

(iv) What conclusion can be drawn from the reversal of the needle's deflection?

20. A crane electromagnet lifts a 500 kg iron load. Explain how the strength can be increased and the risk if the current fails. 5

OR

Discuss how overloading, short circuits and faulty wiring cause fires and suggest five precautions a family should follow. 5

END OF QUESTION PAPER — Total: 40 Marks